
End-to-End Machine Learning Regression System for Institutional Energy Usage Prediction

Project Id: 23

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1. Problem Description

The Student Energy Usage Prediction System is a Python-based machine learning project developed to predict energy consumption values from a cleaned dataset using supervised learning models. The implementation in the training notebook loads `cleaneddataset.csv`, removes non-predictive columns such as `Timestamp`, `Date`, `EnergyUsagekWh`, and `Month`, and uses the remaining variables as input features while `EnergyUsagekWh` is treated as the target variable. The project applies preprocessing, train-test splitting, feature scaling, model training, evaluation, comparison, and visualization in order to support accurate data-driven prediction.

This work reduces manual analysis by automatically learning patterns from historical data and estimating future energy usage values. Such a system can help in understanding usage behavior, improving planning, and supporting efficient monitoring of energy-related trends through machine learning techniques.

1.1 Dataset Description

The dataset used in this project is a cleaned dataset named “`cleaneddataset.csv`”, which contains information related to institutional energy usage.

The dataset includes multiple numerical features such as:

- Time-related attributes (Hour, Day, etc.)
- Environmental conditions (Temperature, Humidity)
- Usage-related parameters (Appliance usage, number of occupants, etc.)

The target variable is:

- `EnergyUsagekWh` – which represents the amount of energy consumed.

Before model training, the dataset is preprocessed by removing irrelevant columns such as `Timestamp`, `Date`, and `Month` to improve model performance. The dataset is structured and suitable for regression-based machine learning models.

1.2 Flow Diagram

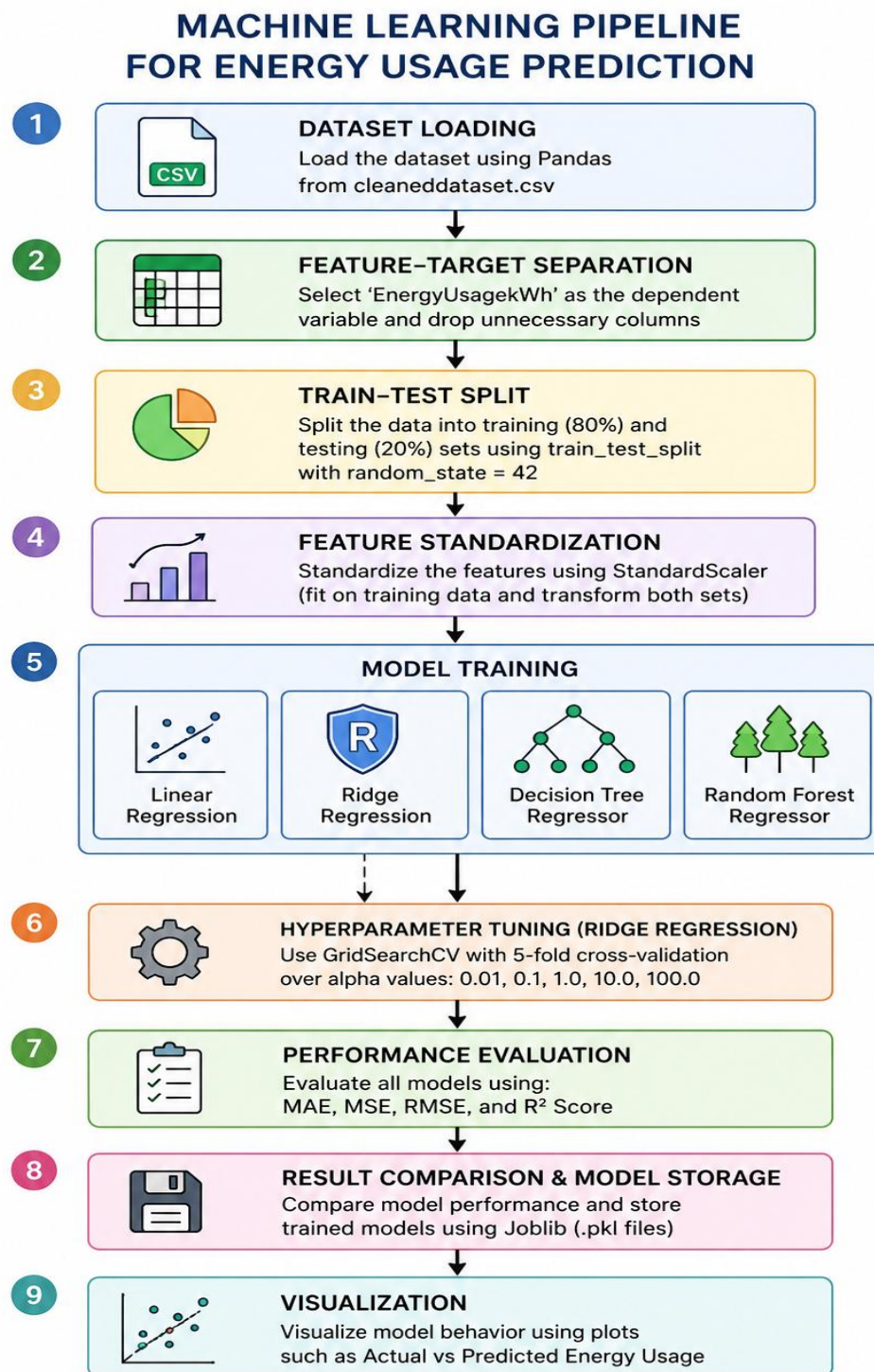


Fig.1 Sample flow diagram.

The flow diagram represents the complete workflow of the machine learning system. It starts with loading the dataset, followed by preprocessing and feature selection. The data is then split into training and testing sets, and feature scaling is applied. Multiple machine learning models are trained and evaluated. Finally, results are visualized and predictions are generated.

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

1. Pandas – Used for data manipulation and analysis.

Functions used:

- `pd.read_csv()` – To load dataset
- `df.head()` – To view data

2. NumPy – Used for numerical computations.

Functions used:

- `np.sqrt()` – For RMSE calculation

3. Matplotlib – Used for plotting graphs and visualizations.

Functions used:

- `plt.scatter()` – For plotting actual vs predicted

4. Seaborn – Used for advanced visualization.

Functions used:

- `sns.heatmap()` – For correlation analysis

5. Scikit-learn – Core machine learning library.

Modules used:

- `train_test_split` – Data splitting
- `StandardScaler` – Feature scaling
- `LinearRegression` – Model training
- `RandomForestRegressor` – Advanced model
- `GridSearchCV` – Hyperparameter tuning
- `metrics` – Model evaluation

6. Joblib – Used to save trained models for reuse.

2.2 Platform Used: (mention all the platforms with the reason of using it)

1. Pandas – Used for data manipulation and analysis.

Functions used:

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- `df.head()` – To view data

2. NumPy – Used for numerical computations.

Functions used:

- `np.sqrt()` – For RMSE calculation

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- plt.scatter() – For plotting actual vs predicted
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Modules used:

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3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

1. Linear Regression

Linear Regression is a supervised machine learning algorithm used for predicting continuous values. It establishes a linear relationship between independent variables (features) and the dependent variable (target).

- **It assumes a linear relationship between input and output.**
- **It is simple, fast, and easy to interpret.**
- **Used as a baseline model in this project.**

2.Random Forest Regressor

Random Forest is an ensemble learning technique that uses multiple decision trees to improve prediction accuracy.

- **It combines multiple trees to reduce overfitting.**
- **It can capture complex and non-linear relationships.**
- **It performs better than linear regression in most real-world scenarios.**

3. Hyperparameter Tuning using GridSearchCV

To improve model performance, hyperparameter tuning was applied using GridSearchCV.

- **It tests multiple combinations of parameters.**
- **It selects the best model based on evaluation metrics.**
- **Parameters tuned include:**
 - **Number of estimators**
 - **Maximum depth**
 - **Minimum samples split**

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

1. Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted values.

Formula:

$$\text{MAE} = (1/n) \sum |y - \hat{y}|$$

2. Mean Squared Error (MSE)

MSE calculates the average of squared differences between actual and predicted values.

Formula:

$$\text{MSE} = (1/n) \sum (y - \hat{y})^2$$

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and gives error in original units.

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

4. R² Score (Coefficient of Determination)

R² score measures how well the model explains the variance in the data.

Formula:

$$R^2 = 1 - [\sum (y - \hat{y})^2 / \sum (y - \bar{y})^2]$$

5. Results and Visualizations

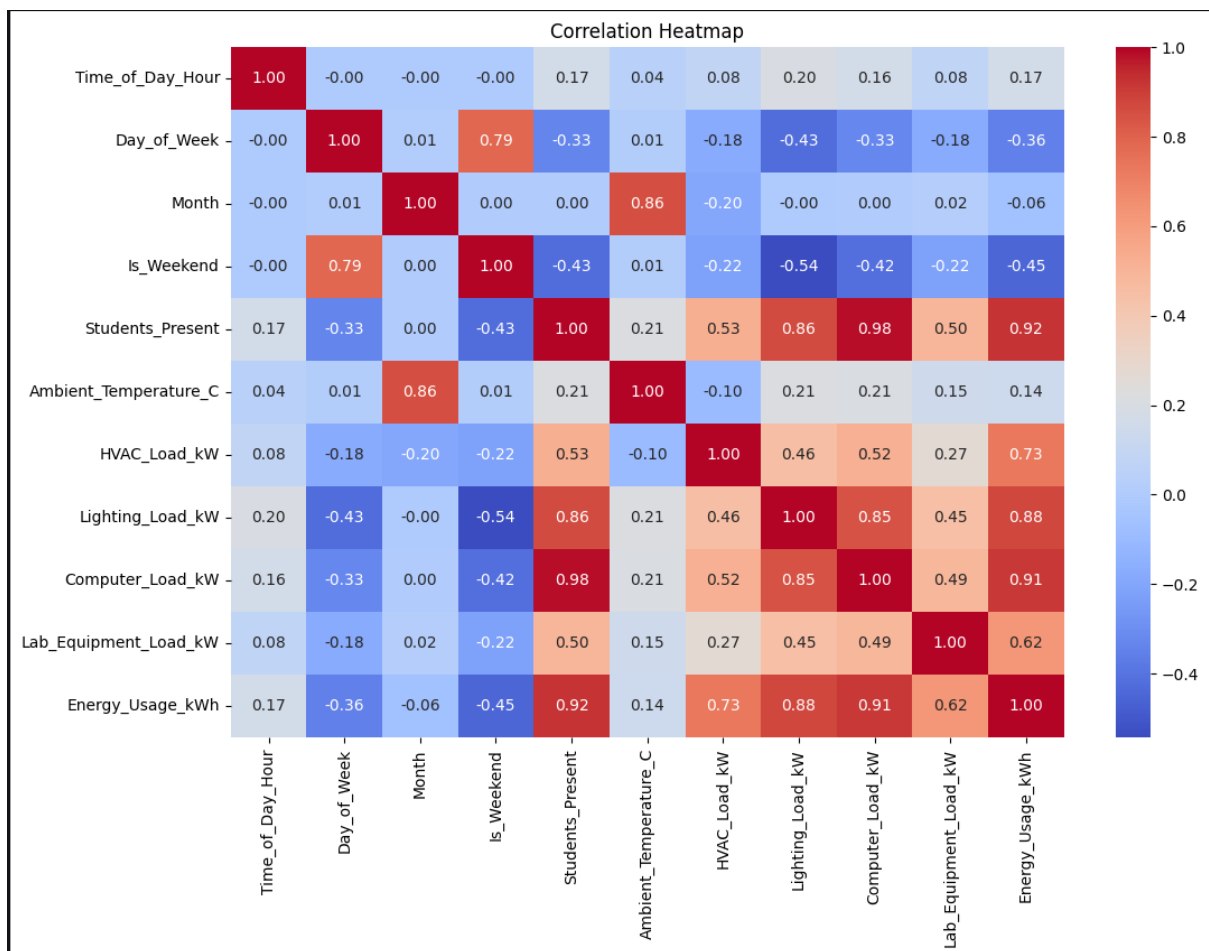


Fig.1Heatmap showing correlation.

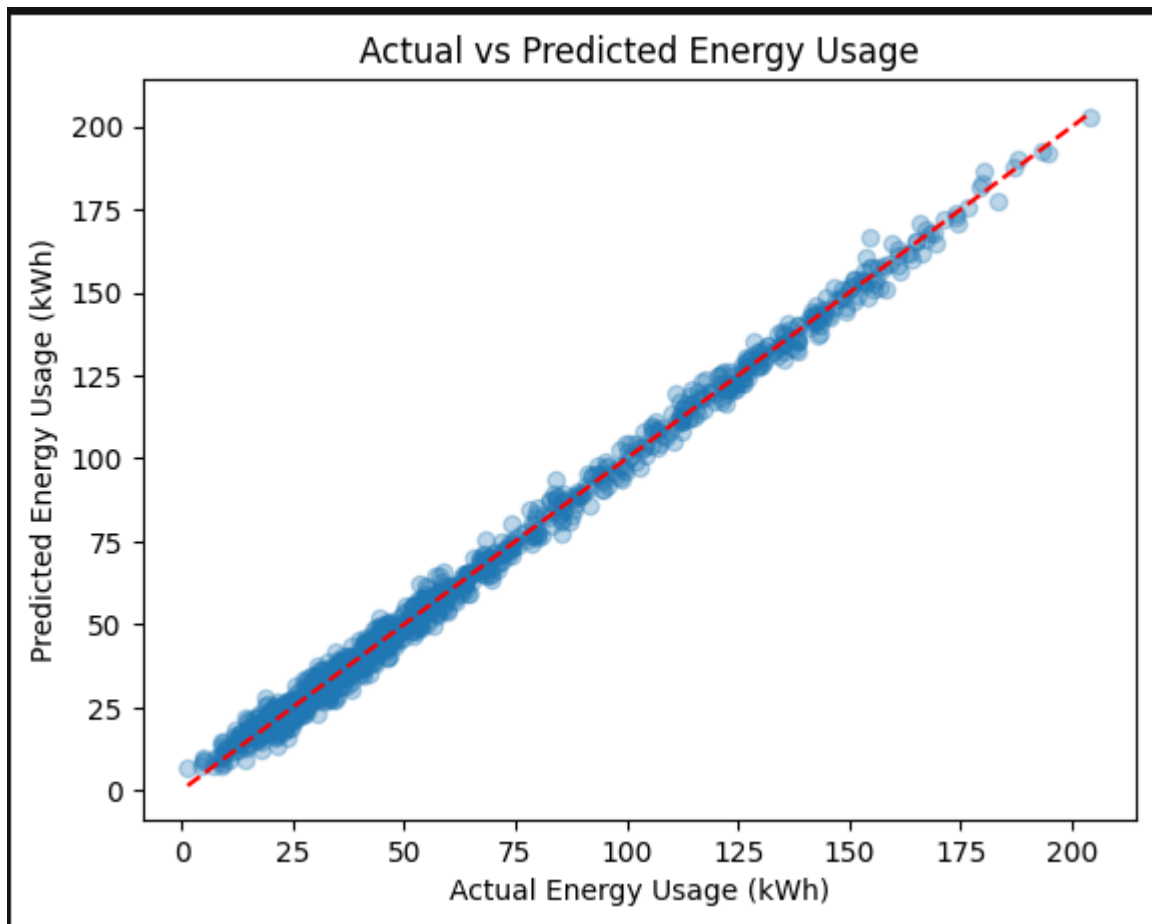


Fig.2 Displays actual vs predicted values.

6. Conclusion and Future Scope:

Conclusion:

This project successfully implemented a machine learning system to predict student performance.

- **Data preprocessing and EDA helped in understanding patterns.**
- **Linear Regression and Random Forest models were trained and evaluated.**
- **Random Forest outperformed Linear Regression.**
- **Hyperparameter tuning further improved model accuracy.**
- **The model can effectively predict outcomes based on input features.**

Future Scope:

- **Use advanced models like XGBoost or Neural Networks.**
- **Deploy the model using web applications (Flask/Streamlit).**
- **Use larger and real-time datasets for better accuracy.**
- **Integrate real-time student monitoring systems.**

7. References:

- Phase 3 base report structure and topic outline from the attached report file.
- Python libraries used in implementation: Pandas, NumPy, Matplotlib, scikit-learn, and Joblib.
- Models and outputs documented in the training notebook, including Linear Regression, Ridge Regression, Decision Tree Regressor, and Random Forest Regressor.

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – <https://www.geeksforgeeks.org/>
- Stack Overflow – <https://stackoverflow.com>

